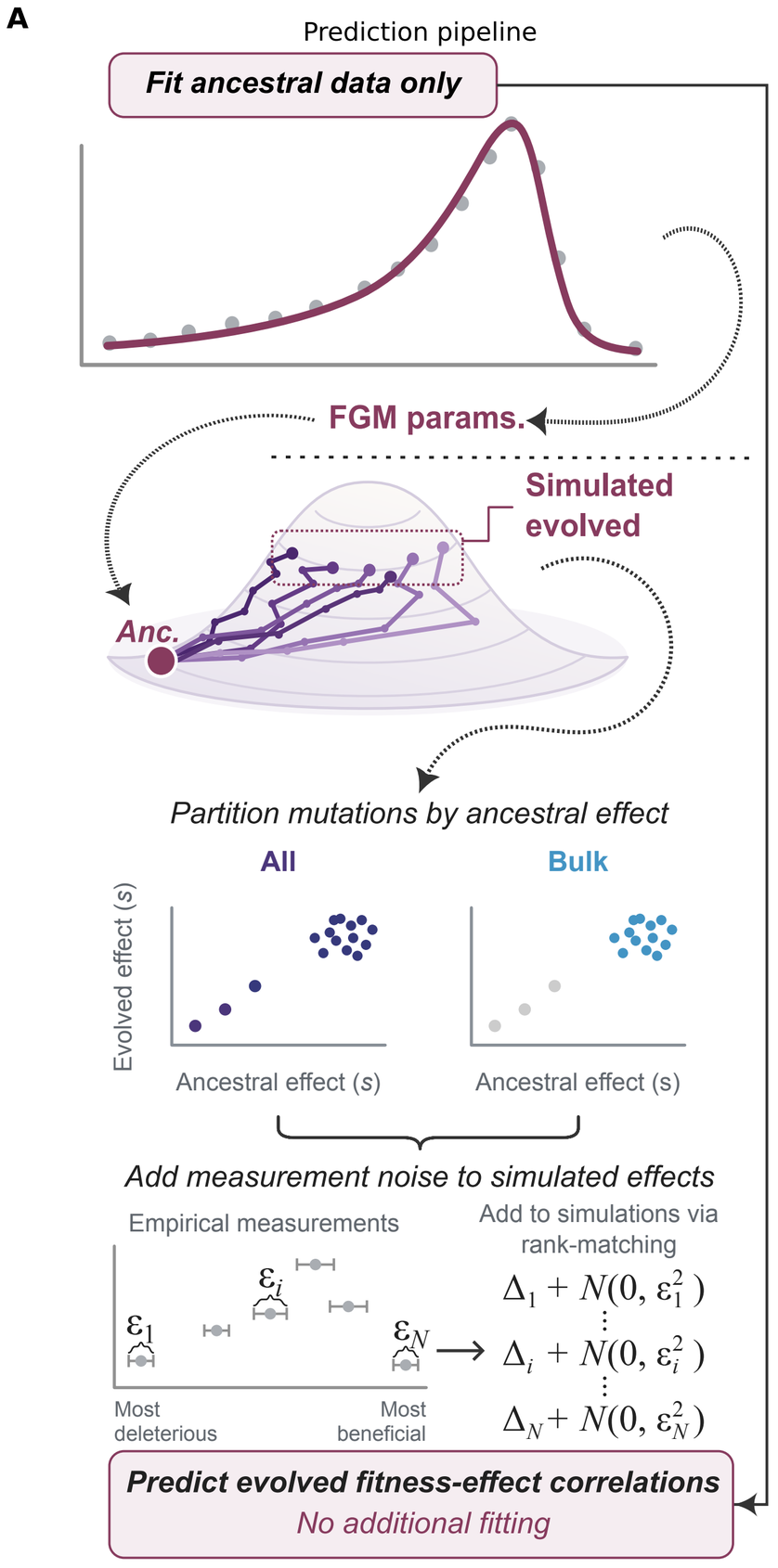
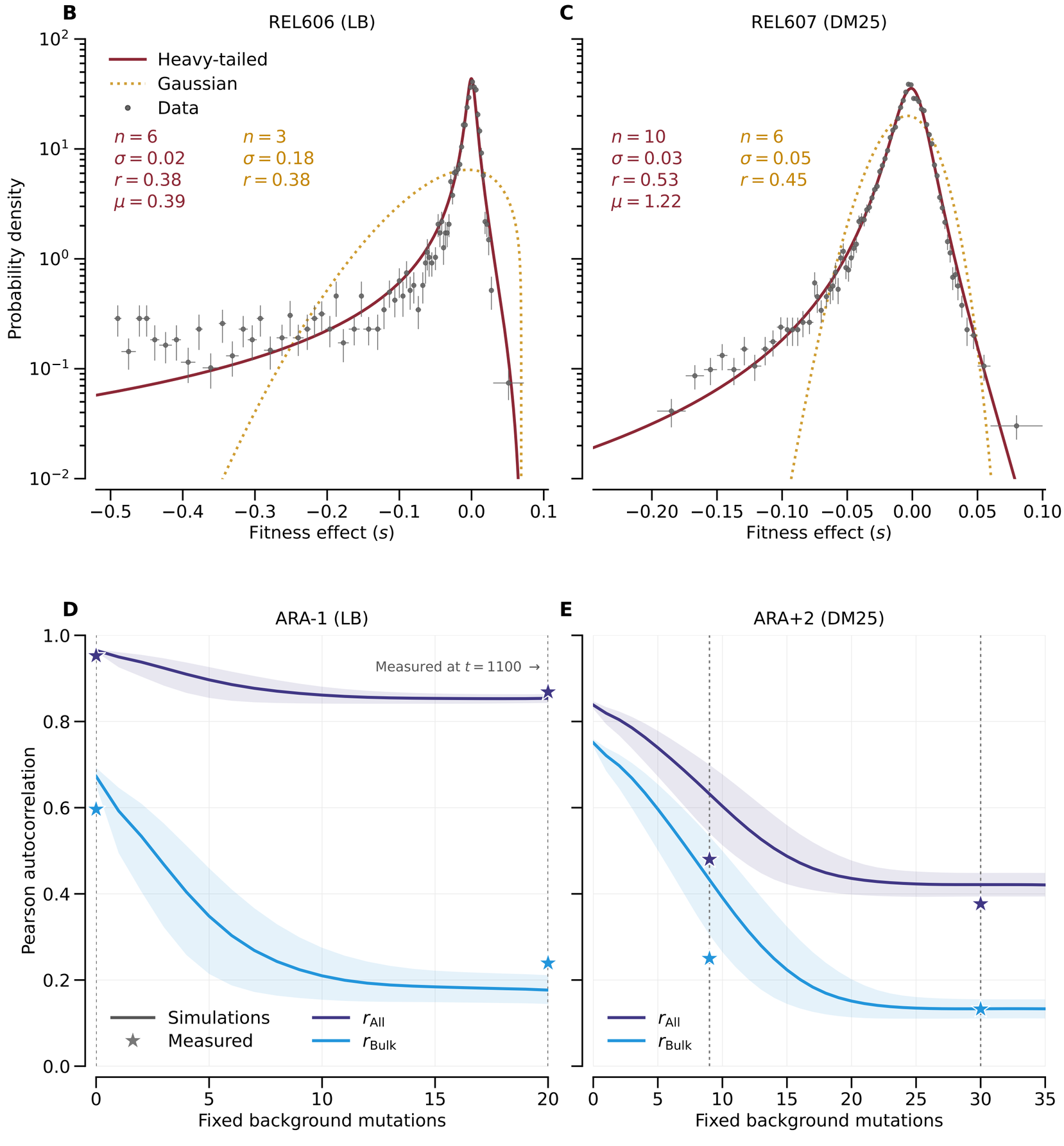

A
Prediction pipeline
Fit ancestral data only
FGM params.
Simulated evolved
Anc.
Partition mutations by ancestral effect
All
Bulk
Evolved effect (s)
Ancestral effect (s)
Ancestral effect (s)
Add measurement noise to simulated effects
Empirical measurements
Add to simulations via rank-matching
ε_1
ε_i
ε_N
$\Delta_1 + N(0, \varepsilon_1^2)$
$\Delta_i + N(0, \varepsilon_i^2)$
$\Delta_N + N(0, \varepsilon_N^2)$
Most deleterious
Most beneficial
Predict evolved fitness-effect correlations
No additional fitting


B
REL606 (LB)
Heavy-tailed
Gaussian
Data
$n = 6$
$\sigma = 0.02$
$r = 0.38$
$\mu = 0.39$
$n = 3$
$\sigma = 0.18$
$r = 0.38$
Probability density
Fitness effect (s)
C
REL607 (DM25)
$n = 10$
$\sigma = 0.03$
$r = 0.53$
$\mu = 1.22$
$n = 6$
$\sigma = 0.05$
$r = 0.45$
Fitness effect (s)
D
ARA-1 (LB)
Measured at $t = 1100$ →
Pearson autocorrelation
Simulations
Measured
r_{All}
r_{Bulk}
Fixed background mutations
E
ARA+2 (DM25)
r_{All}
r_{Bulk}
Fixed background mutations